



## SECTION 1.3 EXERCISES

### Practice Makes Perfect

#### Use Negatives and Opposites of Integers

In the following exercises, order each of the following pairs of numbers, using  $<$  or  $>$ .

185.

- (a)  $9 \underline{\hspace{1cm}} 4$
- (b)  $-3 \underline{\hspace{1cm}} 6$
- (c)  $-8 \underline{\hspace{1cm}} -2$
- (d)  $1 \underline{\hspace{1cm}} -10$

186.

- (a)  $-7 \underline{\hspace{1cm}} 3$
- (b)  $-10 \underline{\hspace{1cm}} -5$
- (c)  $2 \underline{\hspace{1cm}} -6$
- (d)  $8 \underline{\hspace{1cm}} 9$

In the following exercises, find the opposite of each number.

187. (a) 2 (b)  $-6$

188. (a) 9 (b)  $-4$

In the following exercises, simplify.

189.  $-(-4)$

190.  $-(-8)$

191.  $-(-15)$

192.  $-(-11)$

In the following exercises, evaluate.

193.  $-c$  when (a)  $c = 12$  (b)  $c = -12$

194.  $-d$  when  
(a)  $d = 21$   
(b)  $d = -21$

#### Simplify Expressions with Absolute Value

In the following exercises, simplify.

195. (a)  $|-32|$  (b)  $|0|$  (c)  $|16|$

196. (a)  $|0|$   
(b)  $|-40|$  (c)  $|22|$

In the following exercises, fill in  $<$ ,  $>$ , or  $=$  for each of the following pairs of numbers.

197. (a)  $-6 \underline{\hspace{1cm}} |-6|$  (b)  $-|-3| \underline{\hspace{1cm}} -3$

198. (a)  $|-5| \underline{\hspace{1cm}} -|-5|$  (b)  $9 \underline{\hspace{1cm}} -|-9|$

In the following exercises, simplify.

199.  $-(-5)$  and  $-|-5|$

200.  $-|-9|$  and  $-(-9)$

201.  $8|-7|$

202.  $5|-5|$

203.  $|15 - 7| - |14 - 6|$

204.  $|17 - 8| - |13 - 4|$

205.  $18 - |2(8 - 3)|$

206.  $18 - |3(8 - 5)|$

In the following exercises, evaluate.

207.

- (a)  $-|p|$  when  $p = 19$
- (b)  $-|q|$  when  $q = -33$

208.

- (a)  $-|a|$  when  $a = 60$
- (b)  $-|b|$  when  $b = -12$

#### Add Integers

In the following exercises, simplify each expression.

209.  $-21 + (-59)$

210.  $-35 + (-47)$

211.  $48 + (-16)$

212.  $34 + (-19)$

213.  $-14 + (-12) + 4$

214.  $-17 + (-18) + 6$

215.  $135 + (-110) + 83$

216.  $-38 + 27 + (-8) + 12$

217.  $19 + 2(-3 + 8)$

218.  $24 + 3(-5 + 9)$

**Subtract Integers***In the following exercises, simplify.*

219.  $8 - 2$

220.  $-6 - (-4)$

221.  $-5 - 4$

222.  $-7 - 2$

223.  $8 - (-4)$

224.  $7 - (-3)$

225. (a)  $44 - 28$  (b)  $44 + (-28)$

226. (a)  $35 - 16$  (b)  $35 + (-16)$

227. (a)  $27 - (-18)$  (b)  $27 + 18$

228. (a)  $46 - (-37)$  (b)  $46 + 37$

*In the following exercises, simplify each expression.*

229.  $15 - (-12)$

230.  $14 - (-11)$

231.  $48 - 87$

232.  $45 - 69$

233.  $-17 - 42$

234.  $-19 - 46$

235.  $-103 - (-52)$

236.  $-105 - (-68)$

237.  $-45 - (-54)$

238.  $-58 - (-67)$

239.  $8 - 3 - 7$

240.  $9 - 6 - 5$

241.  $-5 - 4 + 7$

242.  $-3 - 8 + 4$

243.  $-14 - (-27) + 9$

244.  $64 + (-17) - 9$

245.  $(2 - 7) - (3 - 8)$

246.  $(1 - 8) - (2 - 9)$

247.  $-(6 - 8) - (2 - 4)$

248.  $-(4 - 5) - (7 - 8)$

249.  $25 - [10 - (3 - 12)]$

250.  $32 - [5 - (15 - 20)]$

251.  $6 \cdot 3 - 4 \cdot 3 - 7 \cdot 2$

252.  $5 \cdot 7 - 8 \cdot 2 - 4 \cdot 9$

253.  $5^2 - 6^2$

254.  $6^2 - 7^2$

**Everyday Math**

255. **Elevation** The highest elevation in the United States is Mount McKinley, Alaska, at 20,320 feet above sea level. The lowest elevation is Death Valley, California, at 282 feet below sea level. Use integers to write the elevation of:
- (a) Mount McKinley. (b) Death Valley.

257. **State budgets** In June, 2011, the state of Pennsylvania estimated it would have a budget surplus of \$540 million. That same month, Texas estimated it would have a budget deficit of \$27 billion. Use integers to write the budget of:
- (a) Pennsylvania.  
(b) Texas.

256. **Extreme temperatures** The highest recorded temperature on Earth was  $57^\circ$  Celsius. The lowest recorded temperature was  $90^\circ$  below  $0^\circ$  Celsius. Use integers to write the:
- (a) highest recorded temperature.  
(b) lowest recorded temperature.

258. **College enrollments** Across the United States, community college enrollment grew by 1,400,000 students from Fall 2007 to Fall 2010. In California, community college enrollment declined by 110,171 students from Fall 2009 to Fall 2010. Use integers to write the change in enrollment:
- (a) in the U.S. from Fall 2007 to Fall 2010.  
(b) in California from Fall 2009 to Fall 2010.